Anil Kumar Karri

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Anil-kumar | AnilKumarKarri

Education

IIIT Srikakulam July 2021 – April 2025
Bachelor of Technology in Computer Science Engineering *CGPA: 8.6/10.0*

IIIT Srikakulam (Intermediate M.P.C) Sept 2019 – Jun 2021
CGPA: 9.49/10.0

Z.P.H School, Chintada (SSC) Jun 2018 – April 2019
CGPA: 10/10

Experience

Python Backend Developer Intern, Algonox Technologies – Hyderabad December 2024 – Present
Tech Stack : Python, Flask, SQL, PostgreSQL, AWS Lambda, EC2, GitLab

- Implemented **Redis-based caching** for the SIM-management dashboard (usage, revenue, SIM's overview metrics), reducing **peak-load** data retrieval time from **8s to 2s** (**75% improvement**) by designing custom caching logic with automated cache invalidation and expiry policies, significantly **boosting system responsiveness** and user experience.
- Automated the end-to-end creation of **10 interdependent modules** by developing an **Excel-based bulk upload system** with parallel processing, cutting manual effort from **60s to 5–7s (90% faster)**. Enabled high-volume ingestion of **10K** records in one run with built-in validation, partial failure handling.
- Developed a **zero-downtime**, multi-threaded sync pipeline migrating **2 lakh records/day** between SQL Server and PostgreSQL with **automated triggers**, retries, and conflict-resolution for consistent cross-system data.
- Built an **automated notifications** workflow that sends **real-time email alerts** for critical events such as recharges, user onboarding, bill generation, device status updates, and SIM changes, improving operational visibility and responsiveness.
- Optimized **Daily carrier data synchronization** by implementing **parallel batch processing** (ThreadPoolExecutor), reducing sync time from **3 hours to 30 minutes** (**80%** improvement compared to **version 1.0**) for **10K** Sims, ensuring the platform remains **up-to-date** and **accurate** for **billing and reporting**, with **retry** handling, and **automated** sync **initiation, success, and failure** notifications.
- Built **Bulk SIM change processing** using **parallel batch execution** (ThreadPoolExecutor), improving throughput from **1 SIM / 2s (v1.0 .NET) to 3–5 SIMs / 2s**, enabling **UI and Excel-based** multi-change-type uploads (customer assignment, rate plan, etc.); integrated **Amazon EventBridge** for time-based scheduling to execute changes on **effective dates**, preventing **billing data mismatches**, with carrier API integration, **retry** logic, **synchronization** with version 1.0 data and consistent updates to **inventory and history** tables.

Technical Skills

Web Technologies: HTML, CSS, Javascript, Reactjs.
Backend: Python, Flask, Django, SQL.
Databases: MySQL, PostgreSQL.
Python/ML Libraries: NumPy, Pandas, Scikit-learn, Matplotlib.
Developer Tools: GitHub, GitLab, Postman, Jira, AWS(Lambda, EC2), GitHub Copilot, VS Code.

Projects

- **AI-Powered YouTube Notes Generator (Flask, NLP, BeautifulSoup):** Built a web app that extracts YouTube transcripts and generates structured summaries using NLP with multilingual support.
- **Intelligent Poly-Culture Irrigation Control System (Flask, NLP, NumPy):** Developed a smart irrigation system that understands natural-language commands to automate watering for multiple crops with accurate plant, bed, and intensity detection using NLP.